

You're Cut Off, Go Home!

Or An Analysis of
Aggravated Assaults
Around Accessible
Alcohol

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The Dangers of Alcohol

- An estimated 88,000 people die every year, according to the Centers for Disease Control and Prevention (CDC), due to alcohol-related issues, including homicide, sexual assault, intimate partner violence, and suicide.
- The American Society of Addiction Medicine notes that between 28% and 43% of violent injuries, and 47% of homicides, alcohol has been estimated to be involved.



Aggravated Assaults Vs. Simple Assaults

What is the difference and why it matters

- Simple assault is an intentional act of violence by another which results in a reasonable fear of harm and minor physical contact
 - Ex. Fist fight, physical bullying, threaten with words or actions
- Aggravated assault is an intentional act of violence **with a weapon** by another which results in a reasonable fear of harm and often major physical contact
 - Ex. Using a gun to shoot someone, a knife to stab someone, throwing household items
- For this project it is important to recognize the difference, as the analysis is only on aggravated assaults

Are Aggravated Assaults More Common Near
Liquor Dispensers?

If so, where?

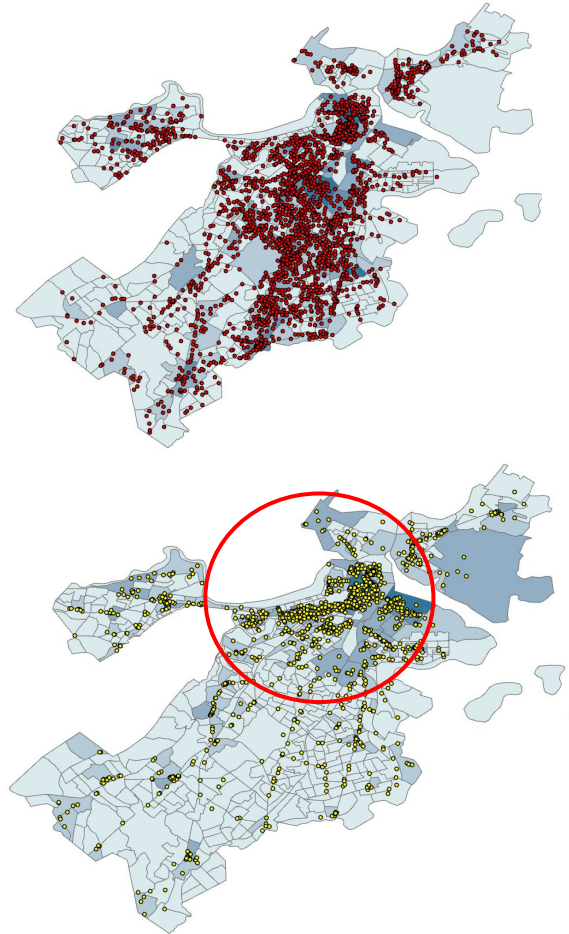
Our Expectations and Limitations

We are hypothesizing that in blocks where more liquor licenses are being held, such as bars and restaurants, individuals will be drinking both on-site and nearby, resulting in higher rates of aggravated assault arrests in that block

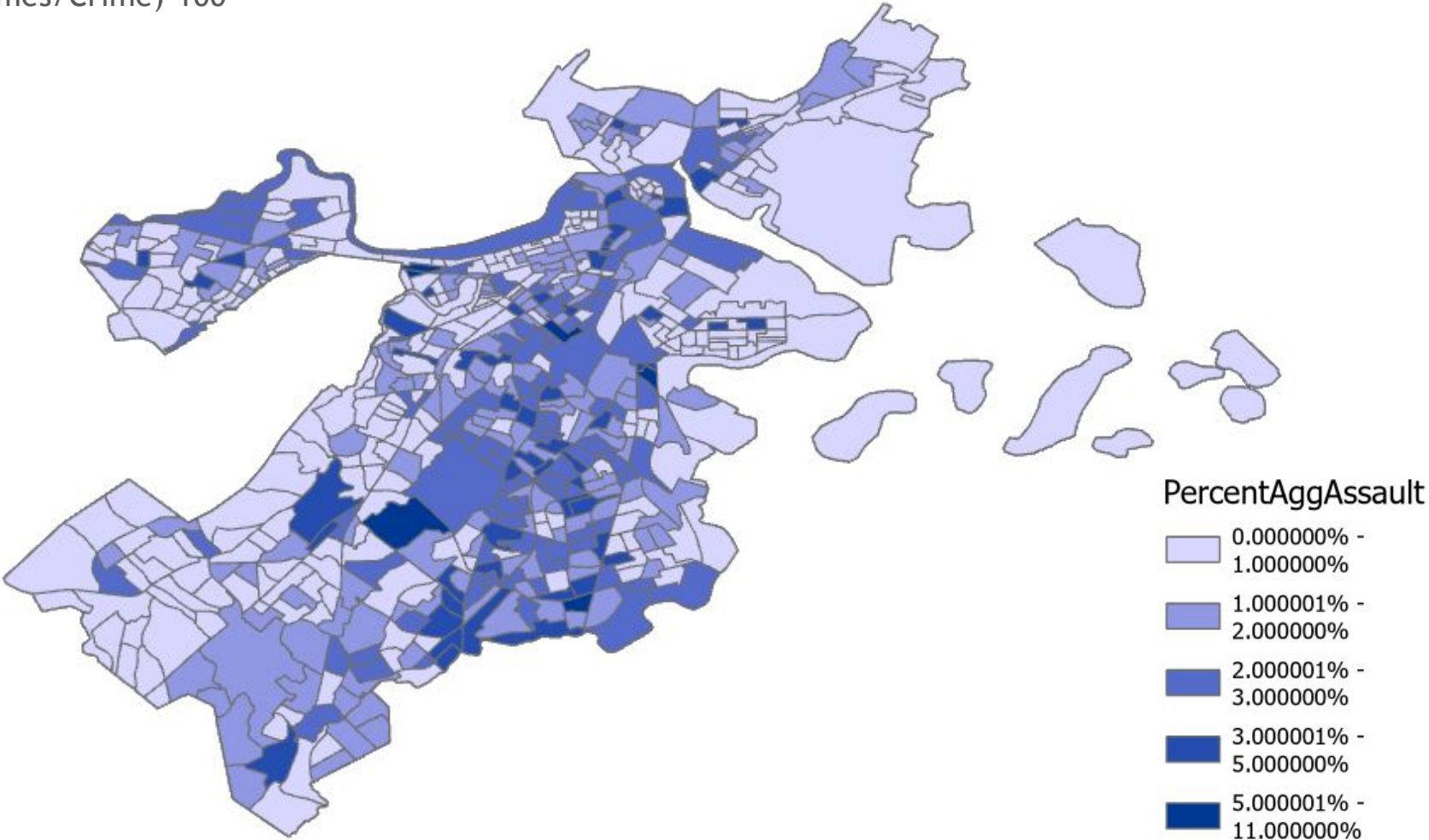
We have limitations in the form of prioritized data collecting in the arrest data. Comorbid charges are not listed, only the main.

Creating Spatial Data

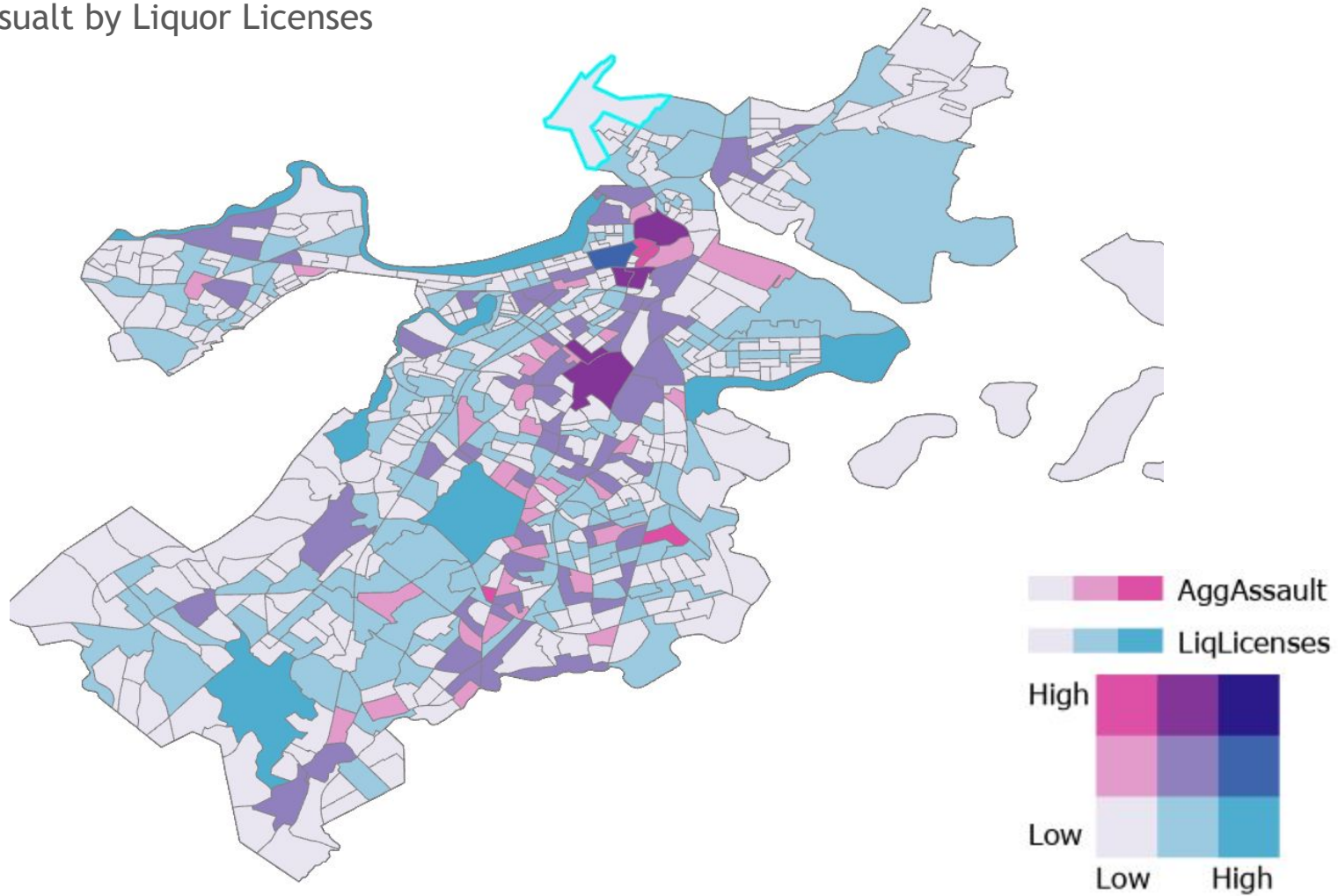
1. Cleaned crime data for Offense Code 423 for Aggravated Assault
 - a. Neither Simple Assault nor Battery were found in the data set
 - b. 5,014 cases were selected out of 155,344
2. Liquor Licenses information download from Boston Government
 - a. Includes 1,544 locations



(Assault Crimes/Crime)*100



Aggravated Assault by Liquor Licenses

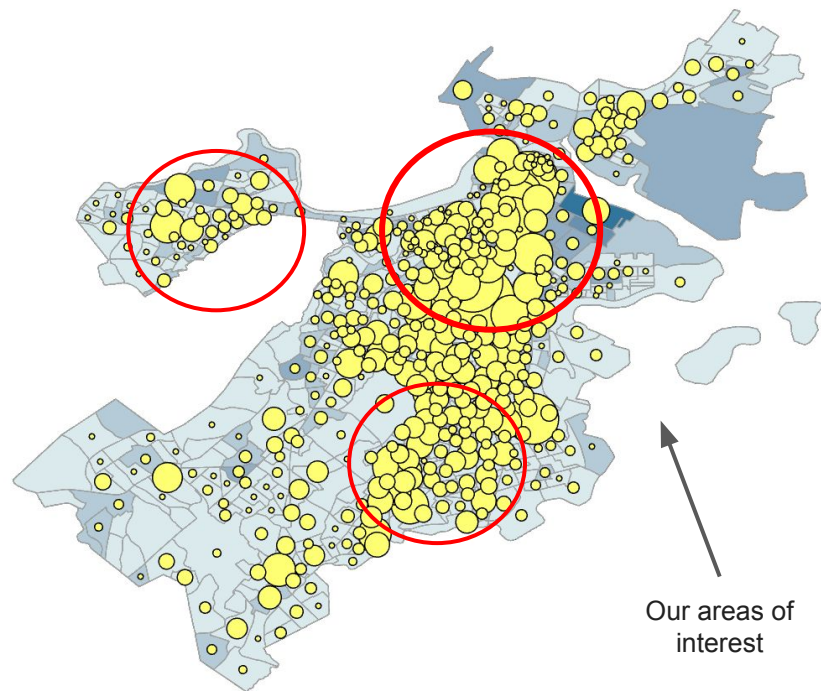
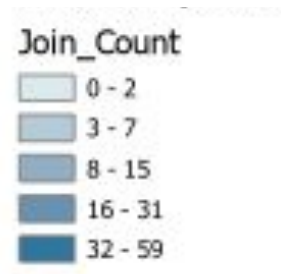
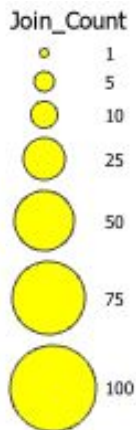


Finding Our Areas Of Interest

Aggravated Assaults are symbolized by proportional symbols

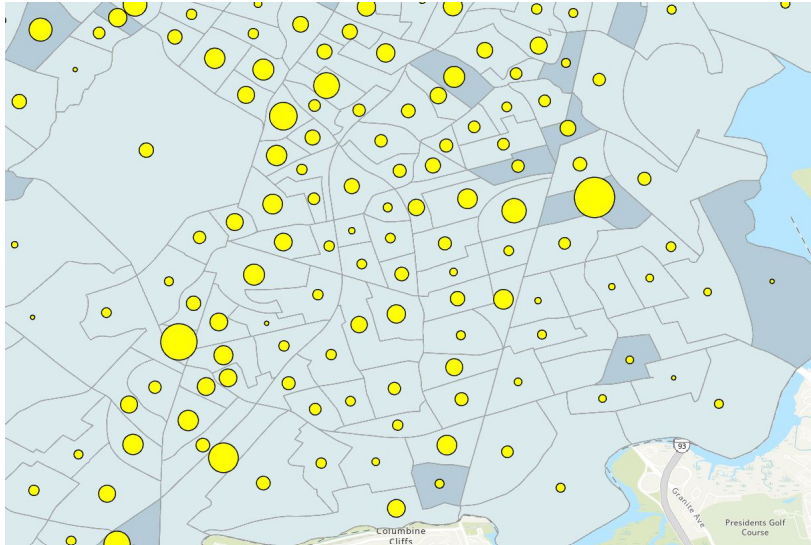
Liquor Licenses are symbolized by graduated colors by the block group

1. Government Center
2. Allston-Brighton
3. Four Corners District

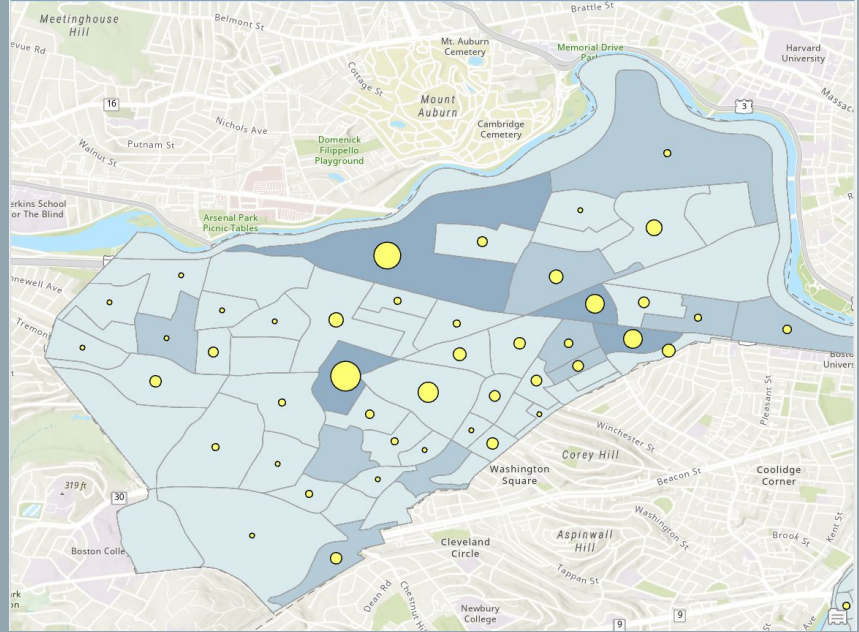


Four Corners District

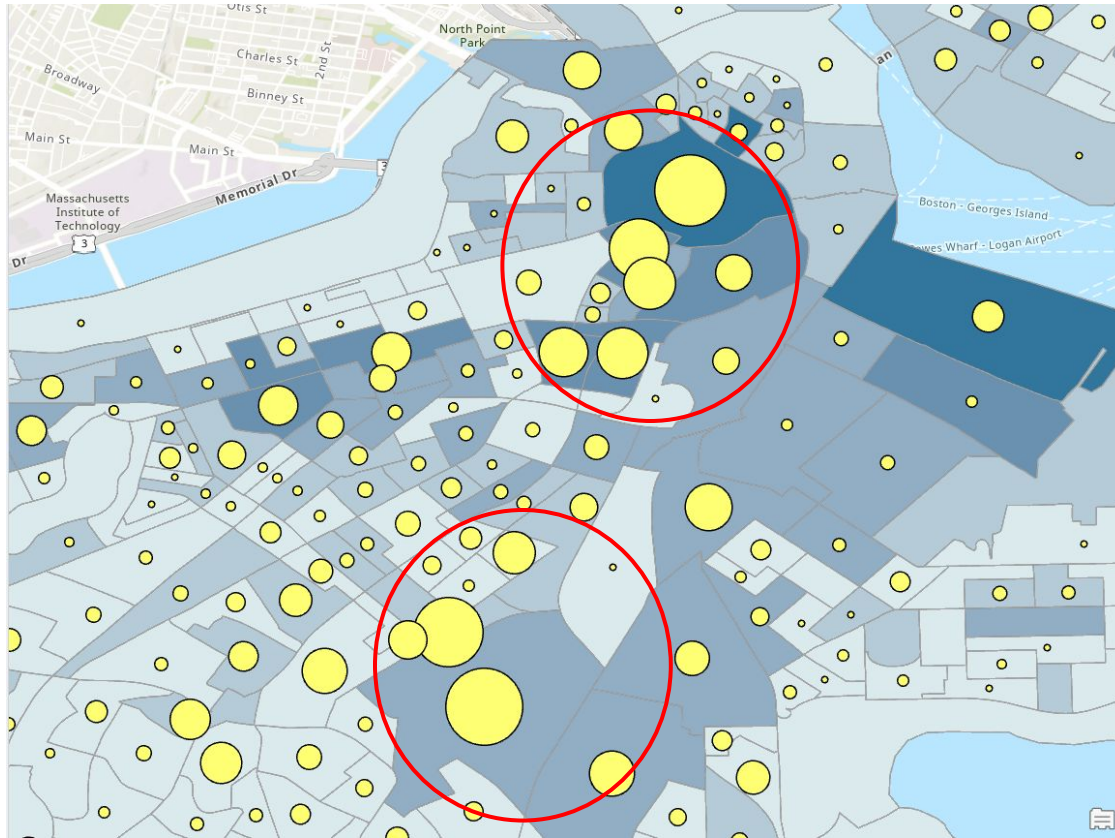
Consistent sizing of yellow circles across consist amounts of liquor licenses per census block



Allston-Brighton



Larger yellow circles indicating more aggravated assaults are in the few darkest census blocks



Government Center

Large areas of
aggravated assaults in
census blocks with
more than 15 liquor
licenses

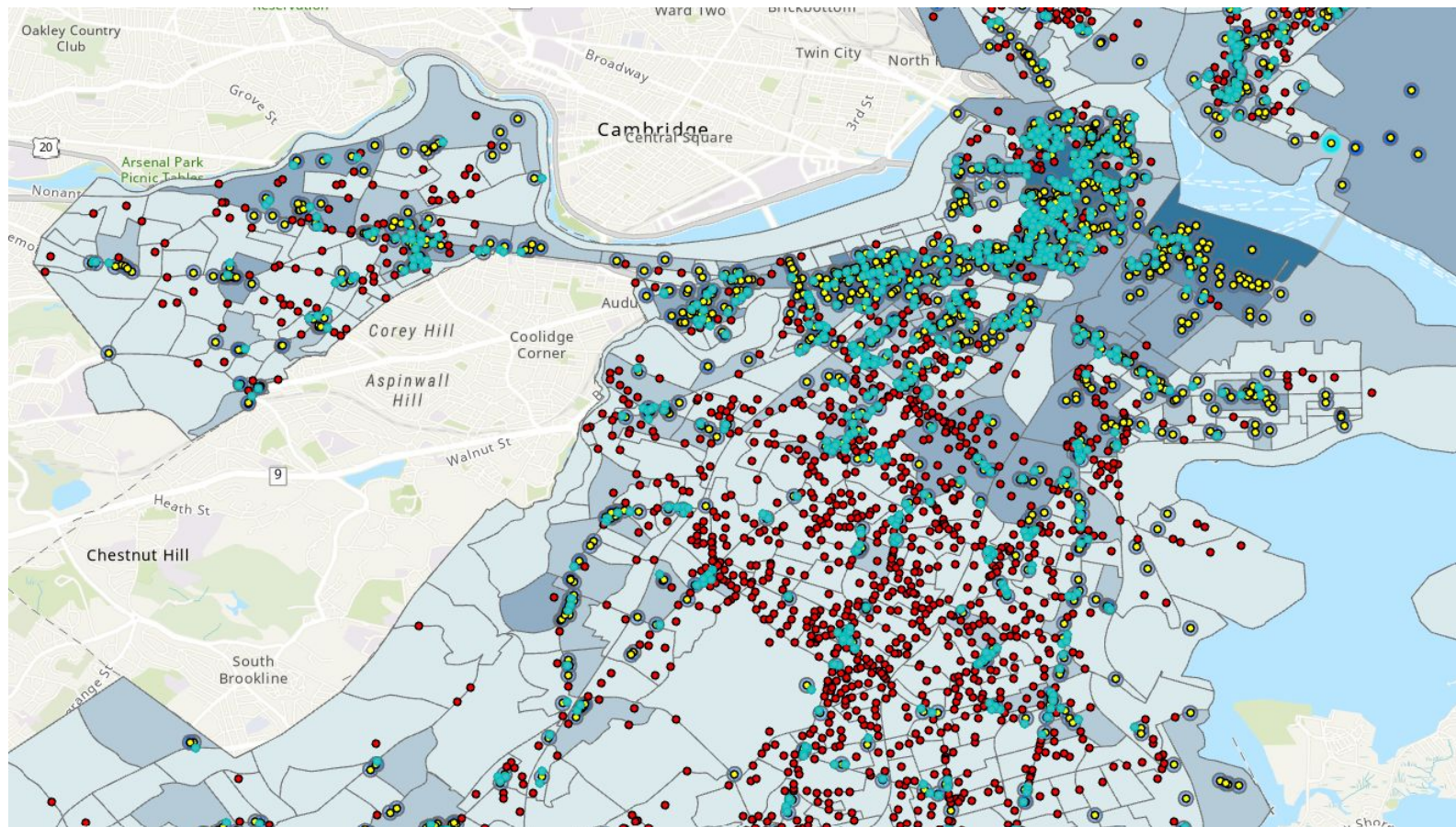
Highly populated area

Nearby public
recreation areas

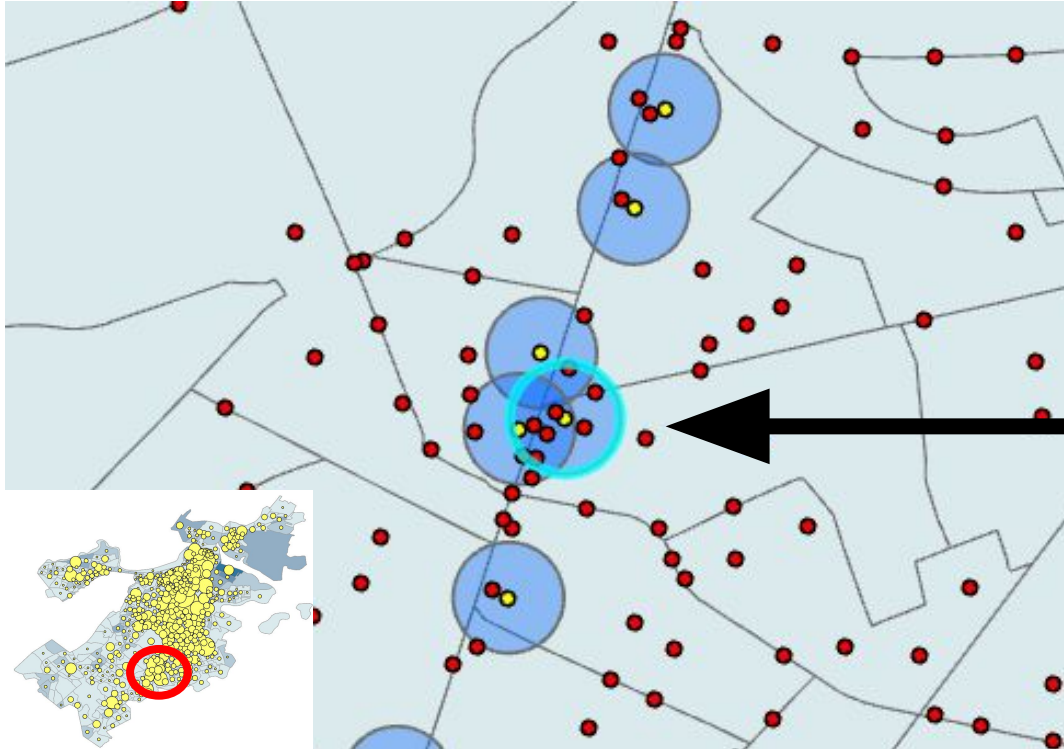
Buffer Analysis of Individual Licenses

- Google estimates a Boston block as about 300 feet
- Using Pairwise Buffers I created buffers within 300 feet of my liquor license points
- Then I created a spatial join between the aggravated assault points and my 300 feet buffers





Where are the Liquor Licenses held that have the most Aggravated Assault within 300 feet?



1. Bon Appetit Restaurant = 69
2. JJDA FERNANDEZ, INC. = 67

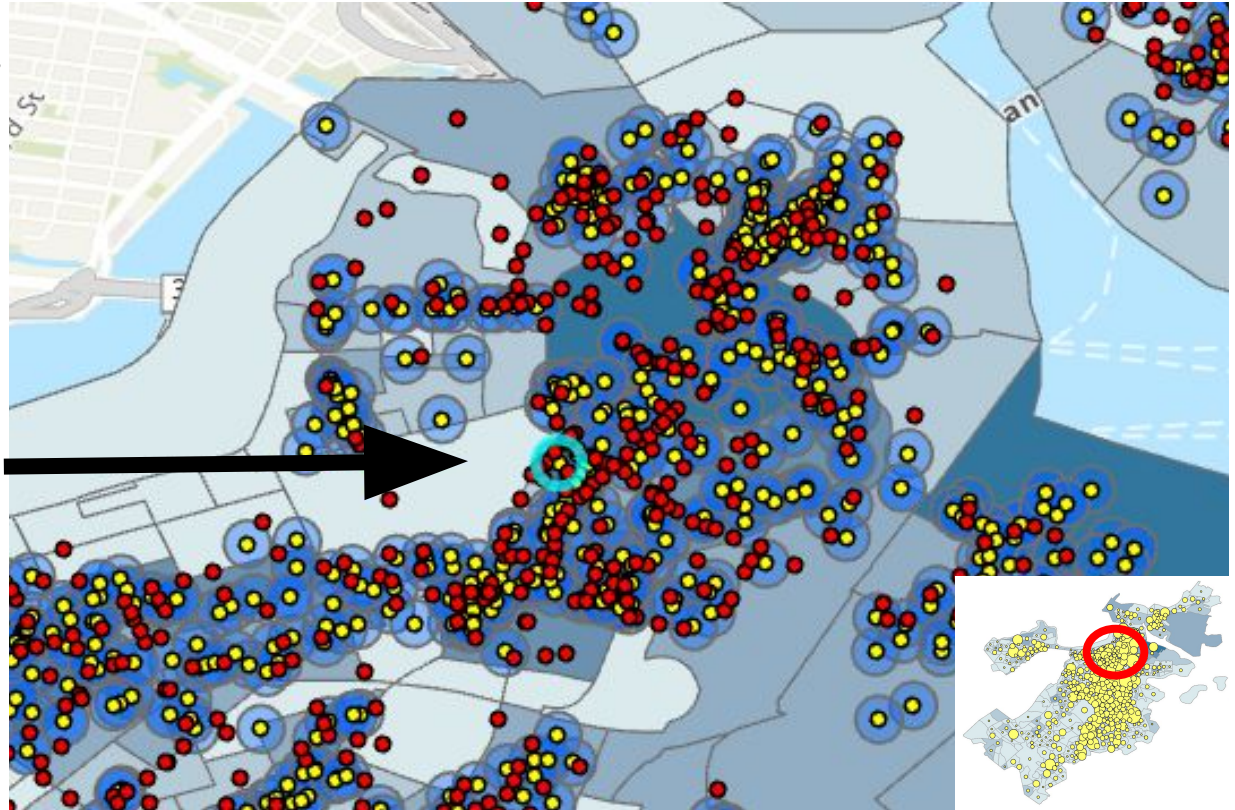


In the Four Corners Area, across the street from each other, are two liquor license locations with more than 60 aggravated assaults since 2023

There is only 3 liquor locations in their combined two block groups

At Number Three, Or Maybe Two?

Epic Marketplace across from
Parkman Plaza of the Commons.
It acts as a general market.
Aggravated Assaults within 300
feet = 47
22 liquor licenses are held
within the census block



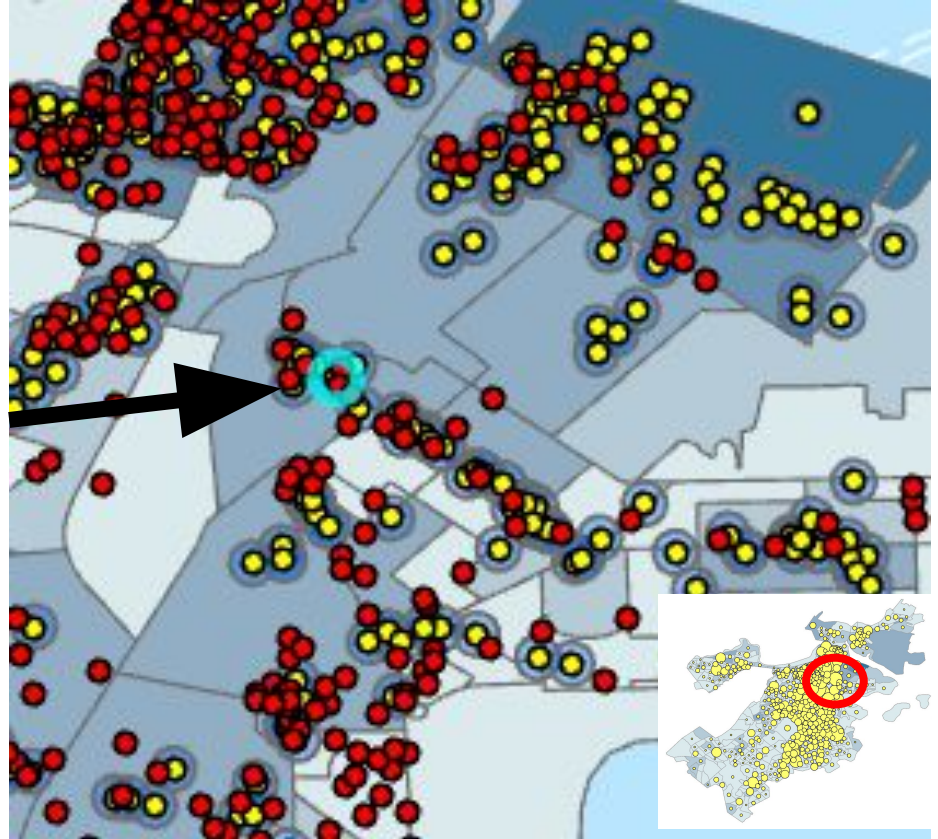
In Fourth And Fifth



4. Amrhein's = 45

5. J&J's Mini Mart = 44

Technically South Boston but in our
'Government Center' area
Both are located in the same
census block with 11 liquor
licenses

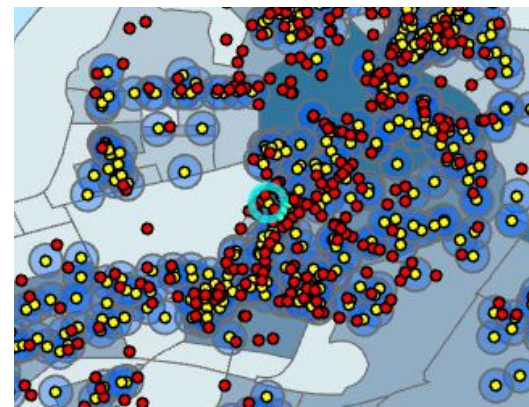
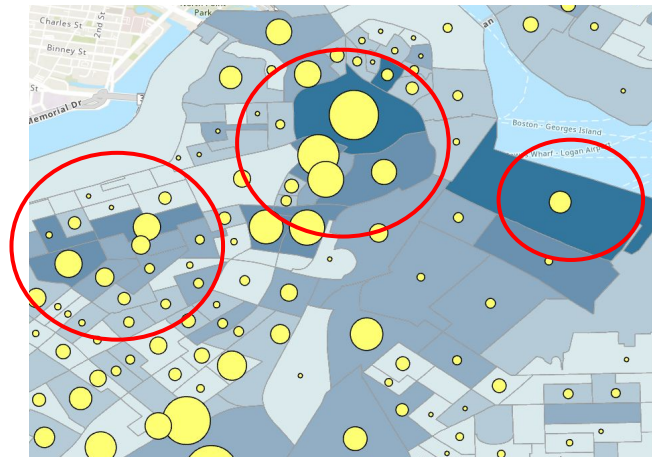


Conclusions Drawn

Our Government Center Area shows us that there tends to be slightly more aggravated assault arrests within census blocks with more liquor licenses

This is highly expected as more liquor licenses are likely to be held within areas with high population density, and more aggravated assaults take place in places of high population density.

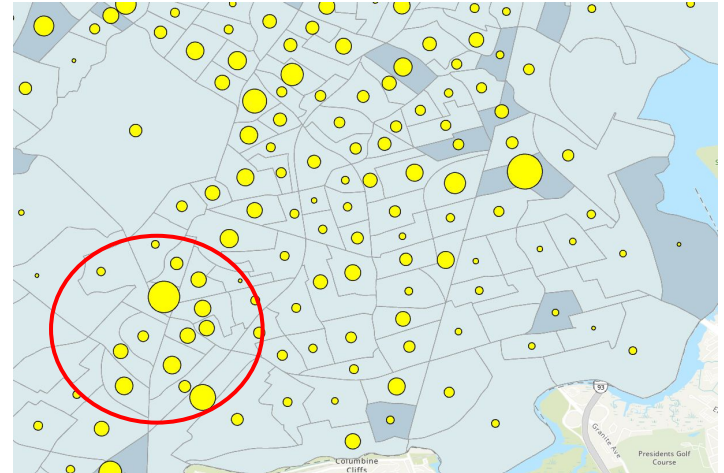
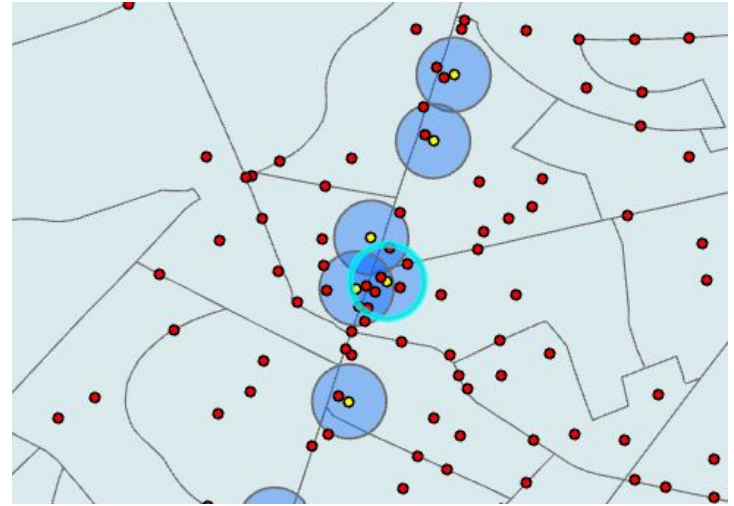
Despite this, there is a dramatic increase from one census block to another centered on darker areas and not *every* area with large amounts of liquor licenses has large amounts of aggravated assaults



Continued...

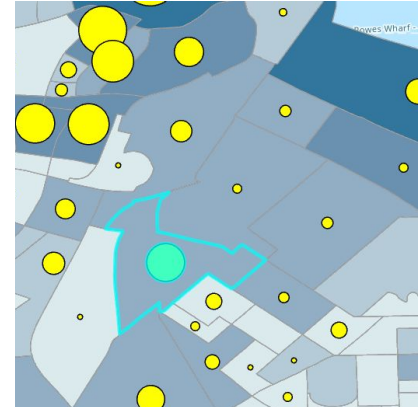
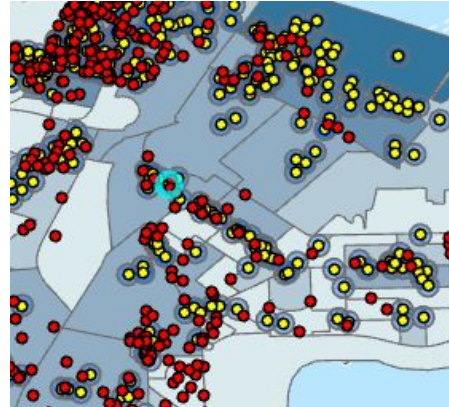
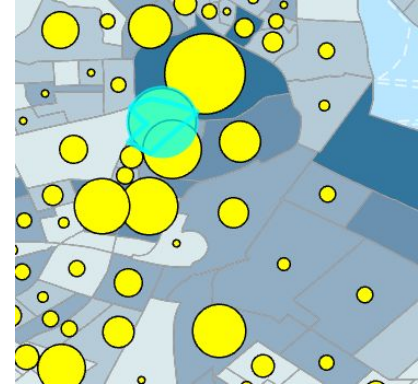
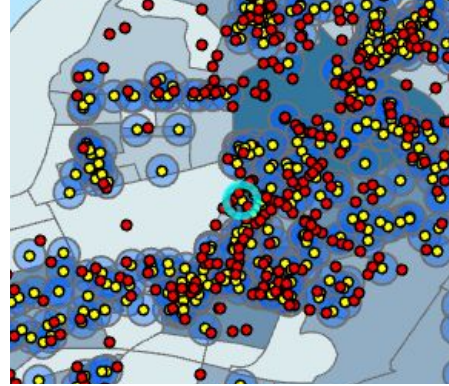
Out of our top 5 liquor license locations of aggravated assaults within 300 feet, the top two were in locations where there are frequent amounts of aggravated assaults within the area and a standard of one or two licenses held per census block

This was to be expected due to the consistency of the area's data



But There's More...

Areas of high aggravated assault and more liquor licenses, the more likely we were to see bars with high amounts of aggravated assaults within 300 feet



In The Future

Future Research:

- Analyze further with additional data that may have comorbid offenses listed
 - Difficult to find with coordinates
- Find information on the relationship between offender and victim
 - Strangers, friends, romantic partners
- Focus more on one type of liquor license

Recommendations:

- Add additional cameras and police surveillance around areas of lots of liquor licenses/high foot traffic
- Keep areas where liquor is sold well lit
- Educate the public on the dangers of public intoxication

Credits

Ackermann, K. (July 25, 2025) The link between violence and alcohol use. *Alcohol.org*. <https://alcohol.org/health-effects/drinking-and-fighting/>

Pridemore, W. A., & Grubestic, T. H. (2011). Alcohol outlets and community levels of interpersonal violence: spatial density, outlet type, and seriousness of assault: spatial density, outlet type, and seriousness of assault. *Journal of Research in Crime and Delinquency*, 50(1), 132-159. <https://doi-org.ezproxy.bu.edu/10.1177/0022427810397952>

Photos from Tripadvisor and Yelp

Crime data for Boston - <https://data.boston.gov/dataset/crime-incident-reports-august-2015-to-date-source-new-system>

<https://data.boston.gov/dataset/crime-incident-reports-august-2015-to-date-source-new-system/resource/3aeccf51-a231-4555-ba21-74572b4c33d6>

Alcohol licenses in Boston - <https://data.boston.gov/dataset/liquor-licenses>

Boston Census groups - <https://data.boston.gov/dataset/census-2020-block-groups>